

CII National Kaizen Circle Compétition 2018

A Kaizen Presentation By

(EVEREST INDUSTRIES LIMITED)

On: 22 & 23rd May-18

Venue : India Habitat Centre , New Delhi

Company Overview:

Everest Industries Limited, incorporated in 1934, has a rich history in manufacturing of Building products and Steel products. Everest offers a complete range of roofing, ceiling, wall, flooring and cladding products distributed through a large network, and also pre-engineered steel buildings for industrial, commercial and residential applications. It is one of the leading building solution providers in India, providing detailed technical assistance in the form of designs, drawings and implementation for every project.

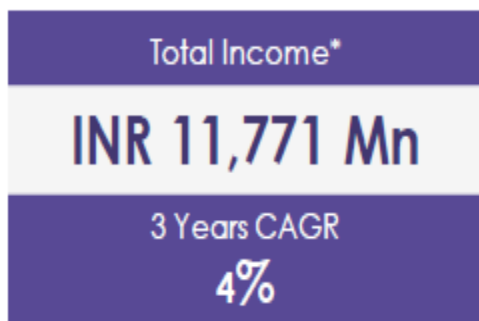
FY17 Business Mix (Revenue Share %):

- **Building products (63%)** – includes fibre cement roofing sheets, fibre cement boards, solid wall panels.
- **Steel buildings (37%)** – offers customised building solutions like Pre-Engineered Steel Buildings, Smart Steel Buildings, metal roofing sheets and cladding.

Manufacturing and Distribution Networks:

- 6 Building Products plants and 3 Steel Building Plants.
- 40 Sales Depot, over 6,000 Dealer Outlets, serving over 600 cities & 100,000 villages.
- Export to over 35 countries (Green solutions – Fibre cement boards).

FY17 Financial Snapshot (Consolidated):



*Total income includes other income

Everest range of products includes



Fibre Cement Roofing
Sheets & Accessories



Hi-Tech Roofing Sheets



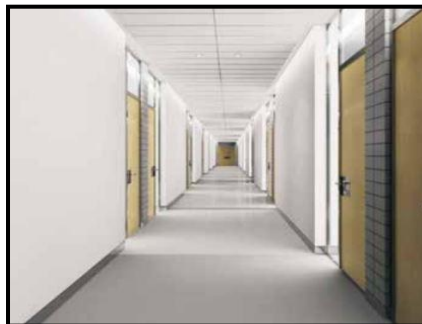
Super Sheets



Cement Planks



Fibre Cement Boards



Rapicon Walls



Smart Steel Buildings



Reliance
Industries Limited



Walmart
Save money. Live better.

IFB

ACC



Mahindra

HYATT
CAPITAL
GATE
ABU DHABI

Pidilite

TATA



Marriott
HOTELS & RESORTS

ITC
ITC Limited

Whirlpool
CORPORATION

RELIANCE

DLF
BUILDING INDIA

NBCC
A Navratna CPSE

Cipla

BRITANNIA
Eat Healthy. Think Better

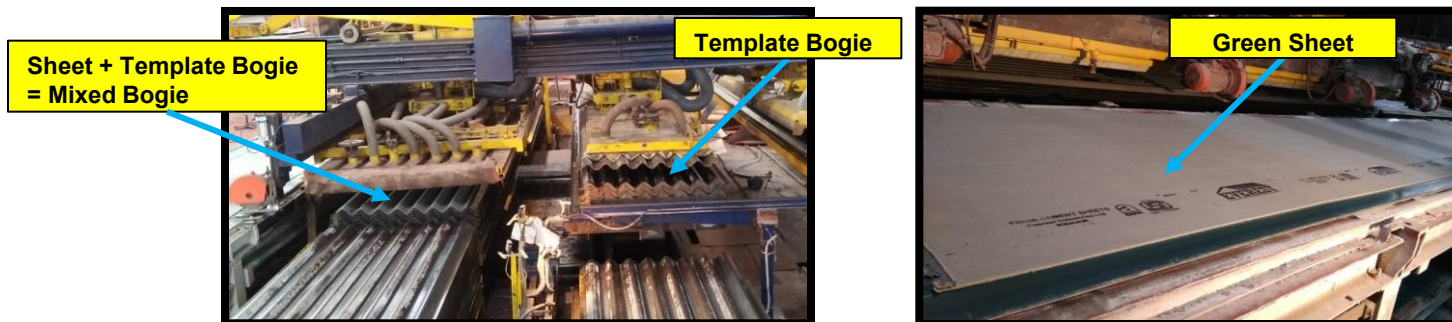
DLF

Coca-Cola

VIDEOCON

Kaizen Theme: Green sheet recovery system during template bogie changeover to avoid Green sheet wastage.

Implemented Area: Sheeting Machine Pilling Area at Podanur Works, Coimbatore



Cross Functional Team:



Mr. M. Vijay Kumar
Electrical Maint.



Mr. K. Senthil Kumar
Electrical Supervisor
Maint.



Mr. Srinivasagam
Electrician



Mr. A. N. Vignesh
Prod.



Mr. N. Anil Kumar
QS & BE



Team Leader
Mr. C. Murugesan
(Sr. Manager, Maint.)



Mr. Devaraj
Fitter



Mr. Suresh
Fitter



Mr. Murali
Fitter

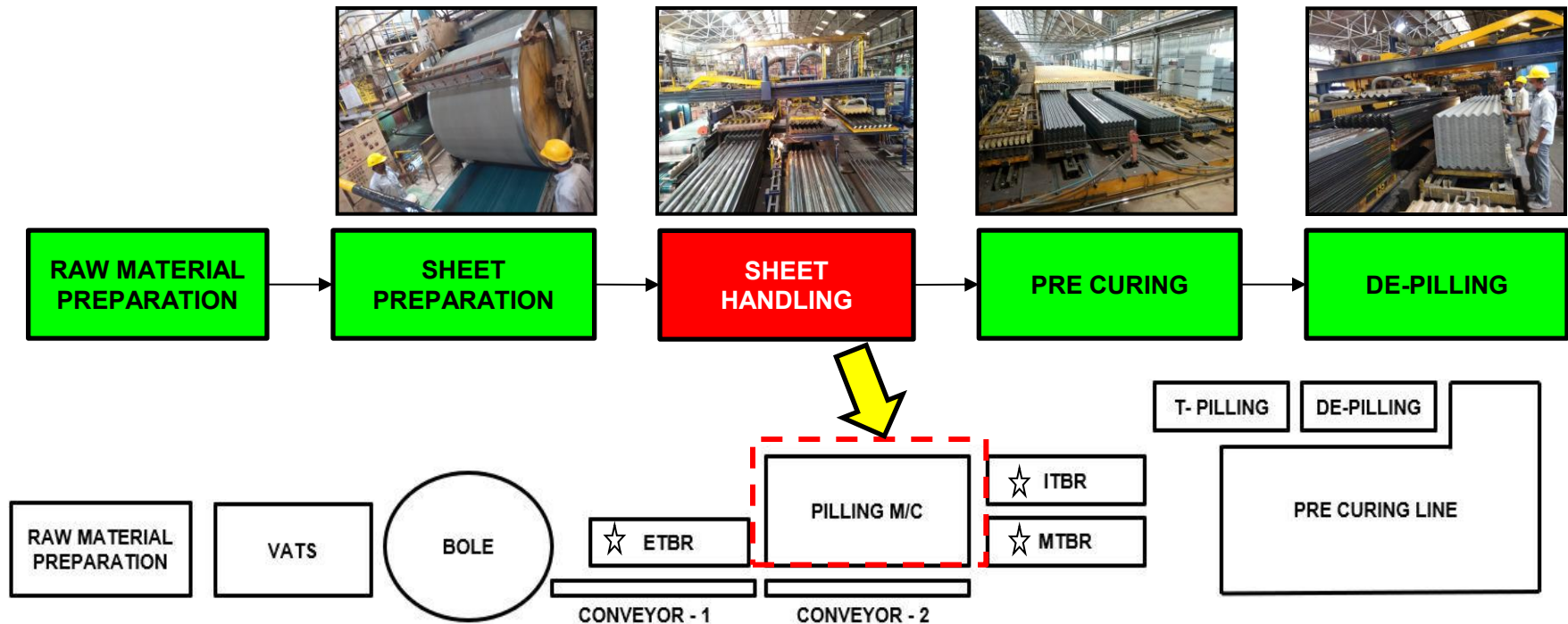


Mr. Pillai
Fabricator



Mr. Shamugaraj
Board Operator

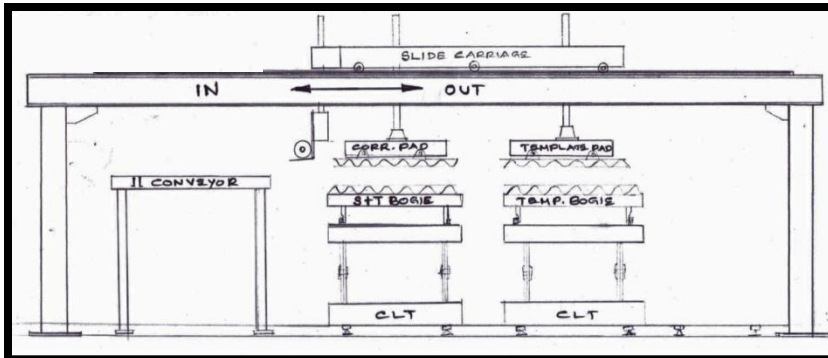
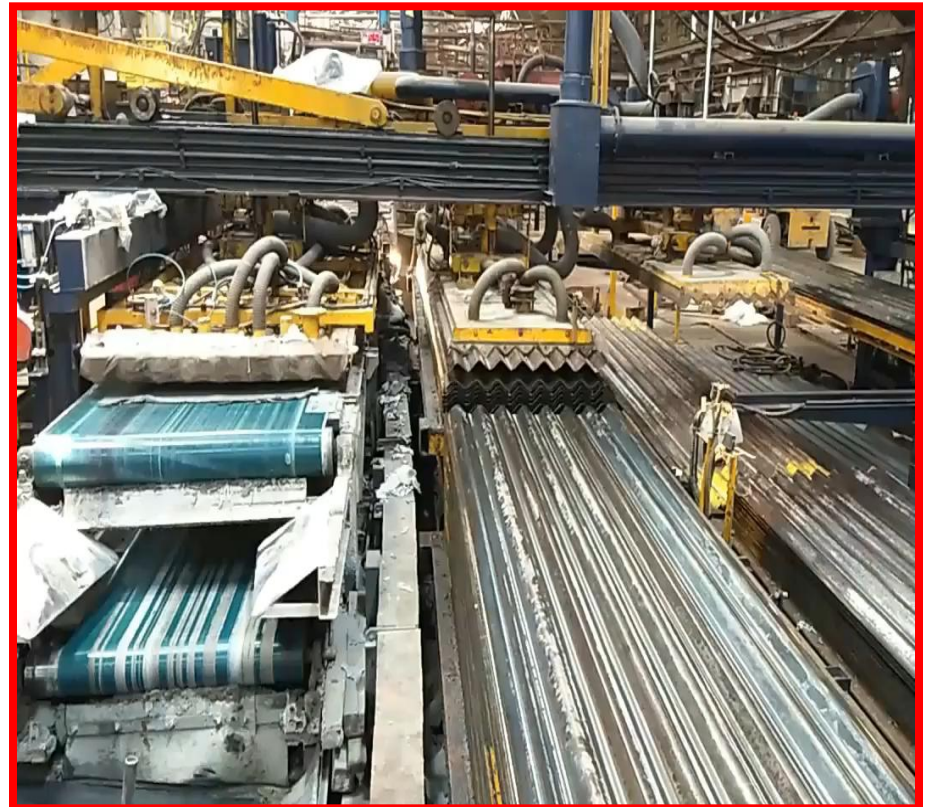
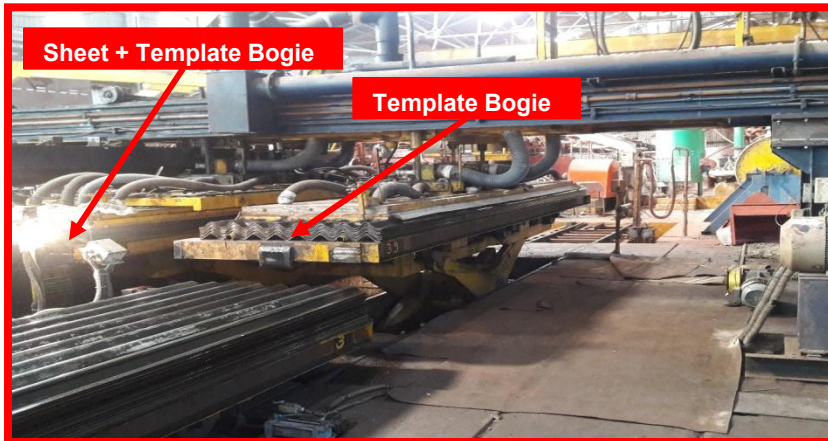
Process Flow:



Due to Non-availability of Metal Templates for Pilling Mixed bogie during Template Bogie changeover there is Loss of Green sheets (2 sheets / Cycle).

- Potential loss of approx. 3.5%
- Approximate Loss = 14 MT/day ~ INR 1.4 Lakh revenue loss/day.

☆ TBR – Transborder Rail



- There was no additional Template pad for recovering templates to stack shuttle trolley with templates.
- There was no side carriage to mount template pilling pad.
- There was no shuttle trolley for stacking recovered templates from template bogie.
- There was no Rail Line & Pneumatic cylinder arrangement for movement of Shuttle trolley IN & OUT.

EVEREST Root Cause Analysis & Idea Generation

Problem: During Metal Template Bogie changeover there is loss of Green sheets due to non availability of Metal Templates.

Why 1 : There are no Templates to provide for mixed bogie pilling during Metal Template bogie changeover.

Why 2 : It takes some time for Metal Template bogie changeover and at that time there is no mechanism for providing templates.

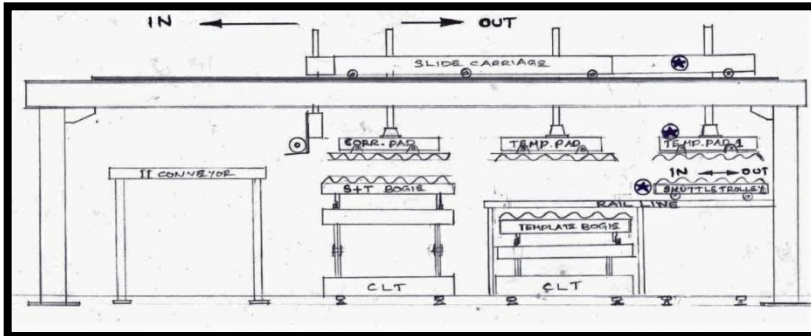
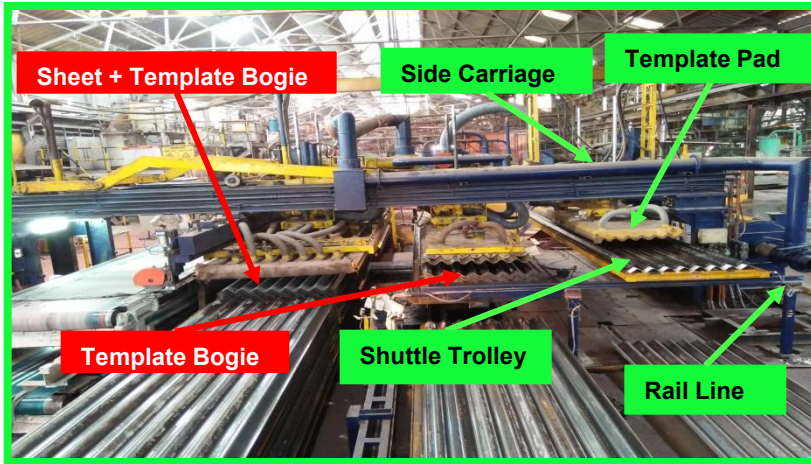
Why 3 : Machine / Design Constraint

Root Cause: No mechanism to supply Metal Templates during template bogie changeover.

Idea Generation: On line supply of additional templates.

Action: Installation of Rolling Shuttle Trolley with additional Template pilling pad to provide templates for Mixed bogie during Metal Template Bogie changeover.

SN	Counter Measure Activities
01	Slide carriage is extended with wheels for mounting of Additional template pad with guide assembly.
02	Additional Template pad is provided with Pneumatic cylinder for collecting templates from template bogie and transfer the same to shuttle trolley to meet the template requirement during bogie changeover.
03	New shuttle trolley is fabricated and provided for template stock and utilizing it during Template Bogie changeover time.
04	Rail line with Pneumatic cylinder is provided for shuttle trolley IN & OUT movement.



- Side carriage is made to mount template pilling pad.
- Template pilling pad is fabricated and installed to side carriage.
- Shuttle trolley is made and installed on Rail Line with Pneumatic cylinder arrangement for movement of Shuttle trolley IN & OUT.
- The entire system is automatized for continuous operation.
- Due to this system there is production saving of 14 MT/day.

Quantitative : Potential Cost Saving of 1.38 Cr/Annum.

Productivity : Maximum Production/day increased from 390 MT to 402 MT.

Quality : Quality of sheets improved due to reduction in recycled waste by 18%.

Cost : Potential Cost Savings = 1.38 Cr/Annum

Safety : The countermeasure is being controlled by PLC. Hence the Idea implemented is safe.

Morale : This Kaizen has been awarded as the Best Kaizen of Quarter 4, 2017-18 in Inter-works competition.



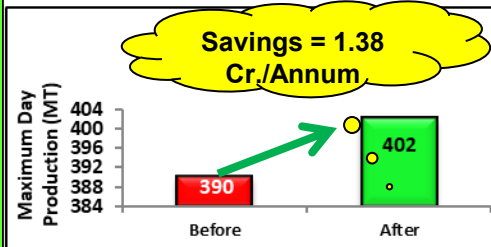
- Training on Standard Operating Procedures are given to all the operators and maintenance people to operate the new system.

Kaizen Sustenance:

WHAT TO DO	Regular Inspection and Maintenance.
HOW TO DO	Regular Training and Standardized Check sheet.
FREQUENCY	Weekly.

Scope of Horizontal Deployment in across Everest :-

Factory	Target Date	Status
1.Bhagwanpur Factory, Roorkee	31.07.2018	Under Implementation
2.Kolkata Factory, Kolkata	30.05.2018	Under Implementation
3.Kymore Factory, Madhya Pradesh	15.06.2018	Under Implementation
4.Podanur Factory, Coimbatore	31.03.2018	Implemented in both the lines
5.Somnathpur Factory, Odisha	30.05.2018	Under Implementation

Kaizen Sheet			Category:	EVEREST	Company	MM/YY	S.No
Reactive	Proactive	Innovative	Productivity Improvement	Everest Industries Ltd	03/18	1	
Kaizen Title: Green Sheet Recovery system during Template bogie changeover to avoid Green sheets wastage.				Implemented Area: Sheeting machine Pilling area			
Problem/Present Status: Loss of Green sheets (2 sheets/ cycle) during Template Bogie changeover. This is a potential loss of approx. 3.5 % which if avoided can lead to productivity improvement. The loss is of approx. 14 MT/day i.e. INR 1.4 lacs revenue loss/day.		Before Improvement: 		Implemented by: Murugesan (TL), Vijay, Senthil, Ajith, Vignesh, Devaraj, Suresh, Murali, Pillai, Shamugaraj, Srinivasagam			
Root Cause Identification: Problem Why 1 Why 2 Why 3 •During Template bogie changeover there is loss of green sheets due to non availability of Templates. •There is no template to provide for mixed bogie pilling during template bogie changeover. •It takes some time for template bogie changeover and at that time there is no mechanism for providing templates. •Machine/Design Constraint		After Improvement: 		Result/Benefit: (a) Qualitative Quality of sheets improved due to reduction in recycled waste by 18%. (b) Quantitative 			
Root cause: No mechanism to supply Templates during template bogie changeover. Idea to eliminate root cause: On line supply of additional templates.		Counter-measure : Additional template storage pad and template pilling pad provided to provide templates to mixed bogie during template bogie changeover.		Standardization: Preventive maintenance as per schedule. Horizontal Deployment: 1. Bhagwanpur Factory -Under Implementation 2. Kolkata Factory - Under Implementation 3. Kymore Factory-Under Implementation 4. Podanur Factory - Implemented 5. Somnathpur Factory-Under			